

# Samples and Chance — Denominators and Biased Samples

## Answer Key

### Grades 6–7

#### Quick Answers

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|-------------------|------------------|------------------------|
| 1. $\frac{5}{12}$ | 4. $\frac{1}{4}$ | 7. 8.5, 3              |
| 2. $\frac{1}{4}$  | 5. 270           | 8. $\frac{3}{10}$ , 90 |
| 3. $\frac{4}{11}$ | 6. 50            |                        |
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#### Curriculum

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**Level:** Grades 6–7

6.SP.B.5 – *problem 7*

7.SP.A – *problems 5, 6, 8*

7.SP.C.5, 7.SP.C.7 – *problems 1, 2, 3, 4*

*Difficulty 1–5 of 5 across 12 tagged checks.*

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#### Common wrong answers

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5. *If they got 18: reported the raw count from the sample as though it already described the whole school*
  6. *If they got 4: counted how many clubs were listed instead of how many students answered*
  7. *If they got 9: took the middle two values as they were written instead of sorting the list first*
  7. *If they got 5.5: took the middle two values as they were written instead of sorting the list first*
  8. *If they got 45: scaled up using the sample size as the school population*
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#### **1. $P(\text{red})$ from a bag of 5 red, 3 blue, 4 green.**

Step 1: fix the denominator before the numerator. “Out of what?” — out of every marble that could be drawn, which is  $5 + 3 + 4 = 12$ .

Step 2: the outcomes we want are the 5 red marbles.

Step 3:  $P(\text{red}) = \boxed{\frac{5}{12}}$ . It does not reduce, since 5 and 12 share no factor.

*Watch for:*  $\frac{5}{7}$ , which compares red with not-red. That is a ratio of red to the rest, not a probability, and it is larger than the true value.

**2. Find and fix: Ines's  $P(\text{blue}) = \frac{3}{9}$ .**

Step 1: name what Ines counted. Her 9 is the number of marbles that are *not* blue (5 red + 4 green). That is the count of the outcomes she did not want — it leaves the blue marbles out of the whole.

Step 2: the denominator must count every possible outcome, blue ones included:  $3 + 9 = 12$ .

Step 3:  $P(\text{blue}) = \boxed{\frac{3}{12} = \frac{1}{4}}$ .

Step 4: a quick sanity check on any probability written this way — the three colour probabilities should add to 1. With Ines's method they come to more than 1, which is impossible.

*Full-credit answer says:* “the denominator counts all 12 marbles, not just the 9 that are not blue.”

**3. Find and fix: Marco after one red marble is removed.**

Step 1: name what changed. Removing a marble and keeping it out changes the *bag*, so both parts of the fraction have to be re-checked — not just the one the question mentions.

Step 2: the numerator is genuinely unchanged: the red marble that left was not green, so there are still 4 green marbles.

Step 3: the denominator did change: 12 marbles minus the one removed leaves 11. So

$P(\text{green}) = \boxed{\frac{4}{11}}$ .

*Watch for:*  $\frac{4}{12}$ , which is Marco's answer, and it is smaller than the truth. Taking a non-green marble out makes green *more* likely, so any answer that goes down is wrong before the arithmetic is even checked.

**4. Find and fix: Max's Year-7 walker.**

Step 1: read who is being chosen. “One student is chosen at random from the class” — so the whole set of possible outcomes is the 24 students, not the 10 walkers.

Step 2: the outcomes we want are the 6 students who both walk and are in Year 7.

Step 3:  $P = \boxed{\frac{6}{24} = \frac{1}{4}}$ .

Step 4: what Max's fraction does mean.  $\frac{6}{10}$  is a correct probability for a *different* question: if a walker is chosen at random, the chance that walker is in Year 7. Same numerator, different whole, different question.

*Full-credit answer says:* “the student was drawn from the class, so the denominator is 24.”

**5. Find and fix: Nora at the bike rack.**

Step 1 (part a): scale the sample proportion up to the school. Of the 20 asked, 18 cycle, so the sample fraction is  $\frac{18}{20}$ . Applying that to 300 students gives  $300 \times \frac{18}{20}$ , which is  $\boxed{270}$ .

Step 2 (part b): read the answer for plausibility. It claims 270 of 300 students cycle, leaving only 30 who walk, ride or are driven — which no school looks like.

Step 3: name the cause. Nora asked at the bike rack, which is a place only cyclists have a reason to stand. Her sample could not have included a non-cyclist, so it was guaranteed to over-state cycling before anyone answered. A fair sample would give every student in the school a chance of being asked — for example by drawing names at random from the roll.

*Watch for:* the answer 18, which repeats the sample count as though it already described the school.

**6. Find and fix: Sam’s library survey.**

Step 1 (part a): add the four columns — the total is the number of *students* who answered, not the number of clubs listed:  $22 + 12 + 10 + 6$ , so 50 students answered.

Step 2 (part b): *flagged for manual review*. Model answer: the survey was taken in the library, where students who prefer quiet indoor activities are already gathered, so chess is over-represented and sport is under-represented. The 50 students are a sample of library users, not a sample of the school, so the result supports “chess is popular among students in the library” and nothing wider.

Step 3: what Sam should do instead. Survey a group in which every student in the school had an equal chance of being asked — names drawn at random from the roll, or one class chosen at random from each year — and ask everyone in that group.

*Watch for:* the answer 4 to part (a), which counts clubs instead of students. A *full-credit part (b)* must name the place as the cause, not just say “the sample is too small” — doubling the number of library students surveyed would not fix anything.

**7. Two samples of reading hours.**

Step 1 (part a): a median needs the values in order. Sample A sorted is 6, 7, 7, 8, 9, 10, 11, 14. With 8 values the median is the mean of the 4th and 5th:  $\frac{8+9}{2}$ , so the median of Sample A is 8.5 hours.

Step 2 (part b): Sample B sorted is 0, 1, 2, 3, 3, 5, 6, 8. The 4th and 5th values are both 3, so the median of Sample B is 3 hours.

Step 3 (part c): the school should use Sample B’s median of 3 hours. Sample A was taken from the book club, a group that reads by definition, so it describes book-club members rather than students in general. Sample B was drawn at random from the roll, so every student had a chance of being included. Sample A’s median is nearly three times as large, which is the size of the error a biased sample can hide.

*Watch for:* 9 for Sample A or 5.5 for Sample B — both come from taking the two middle numbers as printed, without sorting first.

## 8. Show what you know: the later bus.

*Parts (a) and (b) are open — flagged for manual review. Part (c) is checked.*

(a) Model answer: everyone in that queue is already staying late and already using a late bus, so every one of them has a reason to say yes. Students who go straight home — the ones most likely to say no — had no chance at all of being asked. Ben’s estimate will therefore come out *too high*.

(b) Model answer: give every student in the school the same chance of being picked. Draw 150 names at random from the school roll, or take every tenth name from an alphabetical list of all 300 students, and ask everyone drawn. What makes it fair is that the selection is decided before anyone’s answer is known, and it does not depend on where a student happens to be standing.

(c) Step 1: the denominator is everyone who was asked, which is 150, not just the ones who said no. So the fraction is  $\frac{45}{150}$ .

Step 2: divide numerator and denominator by 15:  $P = \boxed{\frac{3}{10}}$ .

Step 3: apply that fraction to the school:  $300 \times \frac{3}{10}$ , so about  $\boxed{90}$  students would use the later bus.

*Watch for:*  $\frac{45}{105}$ , which uses the number who said no as the denominator, and the answer 45, which repeats the sample count instead of scaling it to the school.

*What a full-credit (a) and (b) must contain:* a reason based on *who could have been asked* rather than on sample size, a stated direction for the error, and a plan whose fairness comes from every student having a chance of selection.